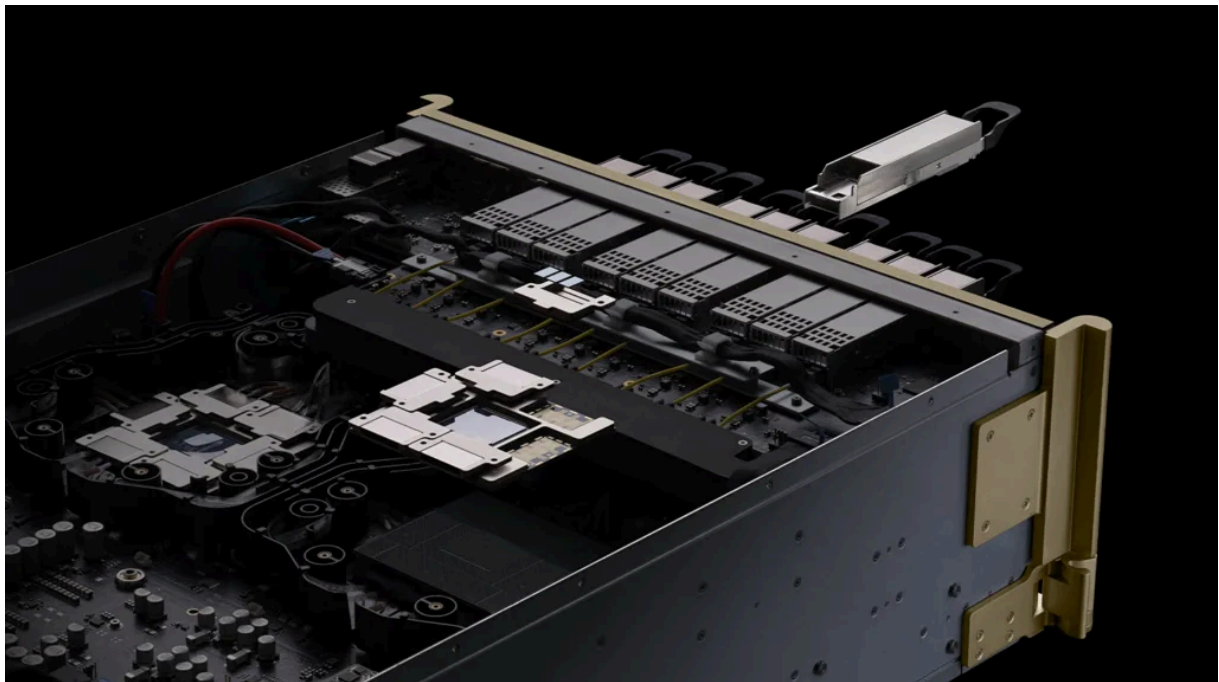




How Industry Collaboration Fosters NVIDIA Co-Packaged Optics



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By [Ashkan Seyed](#)

AI-Generated Summary



NVIDIA is reshaping the landscape of data-center connectivity by seamlessly integrating optical and electrical components. But it's not doing it alone. Central to this breakthrough is close collaboration with partners across the industry.

The NVIDIA networking platform brings together best-in-class technologies from leading partners specializing in advanced foundry processes, cutting-edge lasers, and advanced fiber-to-chip solutions. By coordinating each aspect of the

solution design, NVIDIA leverages not only world-class hardware but also the supporting capabilities essential for realizing scalable, reliable, and high-performance optical systems. The result is a cohesive ecosystem that addresses the growing demands for data throughput and energy-efficient design, setting a new standard for the future of optical interconnects.

Join us as we take an in-depth look at the innovation, partnerships, and technical foundations behind the NVIDIA co-packaged optics (CPO) platform: photonics ICs, electronic ICs, fiber, packaging, connectors, and lasers. By examining these cutting-edge advancements, we'll detail how the NVIDIA ecosystem enables the future of scalable, energy-efficient AI factory interconnects.

How TSMC helped solve Micro Ring Modulator problems

At the heart of CPO innovation lies the NVIDIA Micro Ring Modulator silicon photonics engine—a breakthrough that redefines the possible density and efficiency of optical interconnects. Engineered for direct 200Gbps PAM4 modulation per wavelength, the Micro Ring Modulator achieves an ultra-low density footprint. That enables unprecedented data throughput without sacrificing physical compactness. This level of integration is key for the emerging generation of data-center switches and processors that requires smaller package form-factors to minimize power consumption while maximizing thermal manageability.

Deep collaboration with TSMC has helped NVIDIA solve—at production scale—some of the longstanding manufacturing problems associated with Micro Ring Modulators. That includes precise control of the fabrication process, mitigation of thermal sensitivity, and consistent high-speed modulation. Through advanced process engineering and refined control techniques, NVIDIA and TSMC have achieved reliable, repeatable performance, ensuring that even at the smallest geometries, each modulator meets exacting standards required for hyperscale deployment.

What sets the Micro Ring Modulator apart is its combined small footprint and robust thermal control. This marriage of size and stability upends the traditional

floorplan constraints imposed by bulkier modulator types, unlocking the potential for multi-row scaling of optical interconnect throughput within a single package. As a result, dense arrays of high-speed interconnects become feasible, dramatically increasing bandwidth per device without compromising power or thermal budgets.

How compact, optimized electro-optical engines enable power efficiency

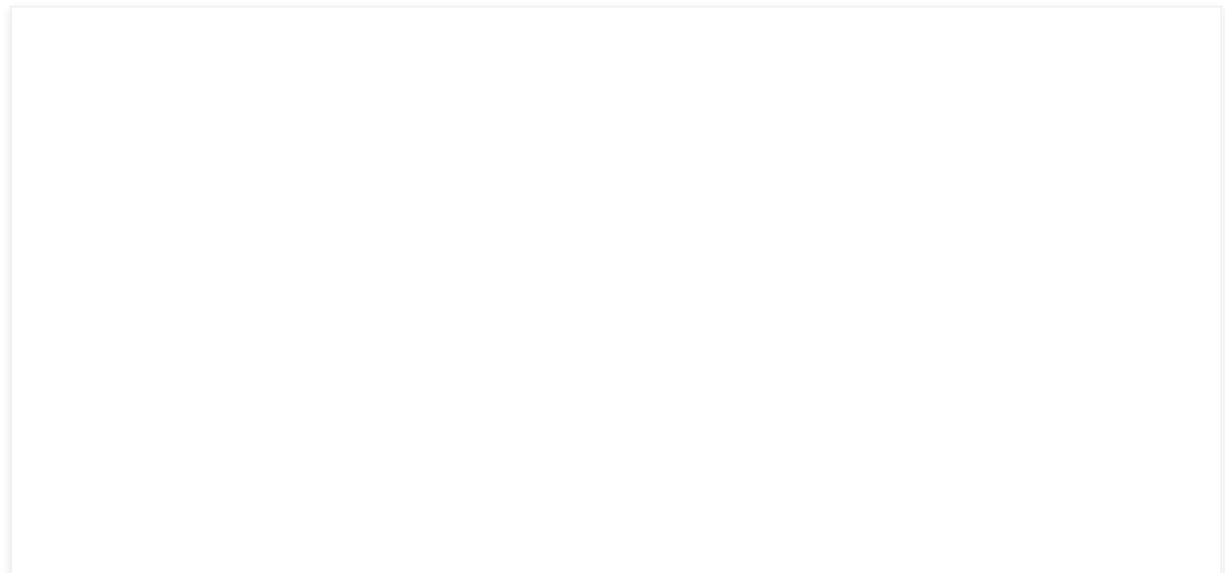
When electro-optical interconnects are engineered in tandem, parasitic losses are minimized. Stacked CMOS and photonics components within mere microns of each other enable ultra-dense and efficient integration. This reduces power consumption and paves the way for compact, thermally manageable modules compatible with hyperscale data center environments.

Wafer-level integration of micro-lenses reduces both the duration and sensitivity of fiber alignment processes. This advancement streamlines production and enhances scalability, making wafer-scale manufacturing of high-performance photonics modules a practical reality.

How modular optical subassemblies ensure rapid deployment

The Quantum-X Photonics switch is powered by a sophisticated optical subassembly designed to deliver unprecedented bandwidth and connectivity for next-generation data centers. Each optical subassembly is engineered to provide

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In parallel, the subassembly maintains a hermetically sealed fiber interface to the optical engines. This airtight, contaminant-resistant design guarantees long-term reliability and performance, even as operational demands grow.

A New Era in Data Center Networking with NVIDIA Silicon Photonics-based Network Switching

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NVIDIA's adoption of high-power ELS modules enables Quantum-X Photonics

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Quantum-X switches 576 transmit lanes. It delivers robust, distributed light to the co-packaged photonics engines and enables the unparalleled bandwidth density that distinguishes the Quantum-X Photonics platform from other offerings. Spectrum-X Ethernet Photonics switches will use the same ELS module, with 16 modules for the single-ASIC and 64 modules for the quad-ASIC versions, respectively.

Video 1. A video depicting an NVIDIA Quantum-X Photonics switch connected to an NVIDIA GB300 NVL72 rack-scale AI supercomputer.

Learn more about the NVIDIA CPO platform

Quantum-X Photonics and Spectrum-X Ethernet Photonics are the result of years of innovation led by NVIDIA in collaboration with its ecosystem partners. These co-packaged switches address complex challenges at the silicon and transistor levels, as well as thermal management, reliability, high-volume fiber integration with the chip, and consistent laser performance.

Quantum-X Photonics and Spectrum-X Ethernet Photonics stand as key pillars, enabling the highly reliable, power-efficient networking infrastructures required by next-generation AI factories. By blending modularity, efficiency, and robust engineering, these platforms ensure data centers can seamlessly scale and meet the exponential demands of AI-driven applications, securing NVIDIA's leadership in the future of data center connectivity, as discussed further [here](#).

For more, [check out the previous CPO technical blog](#) and [our video on the deep dive into co-packaged optics switches](#). And [visit our silicon photonics website](#) for continued announcements and product release updates.

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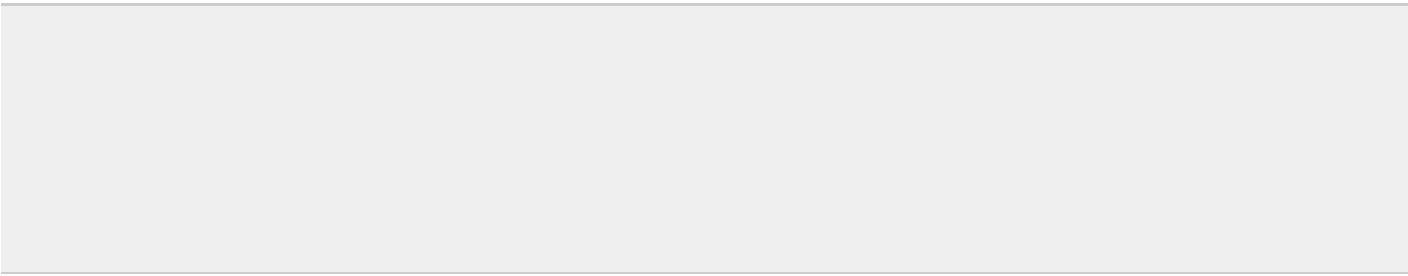
About Ashkan Seyedi

Ashkan Seyedi is a director of product marketing who works on developing high-bandwidth, efficient optical interconnects for AI and high-performance computing. He previously worked at Intel and Hewlett Packard Enterprise. Ashkan received a dual bachelor's in electrical and computer engineering from the University of Missouri in Columbia and a Ph.D. from the University of Southern California, where he worked on photonic crystal

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